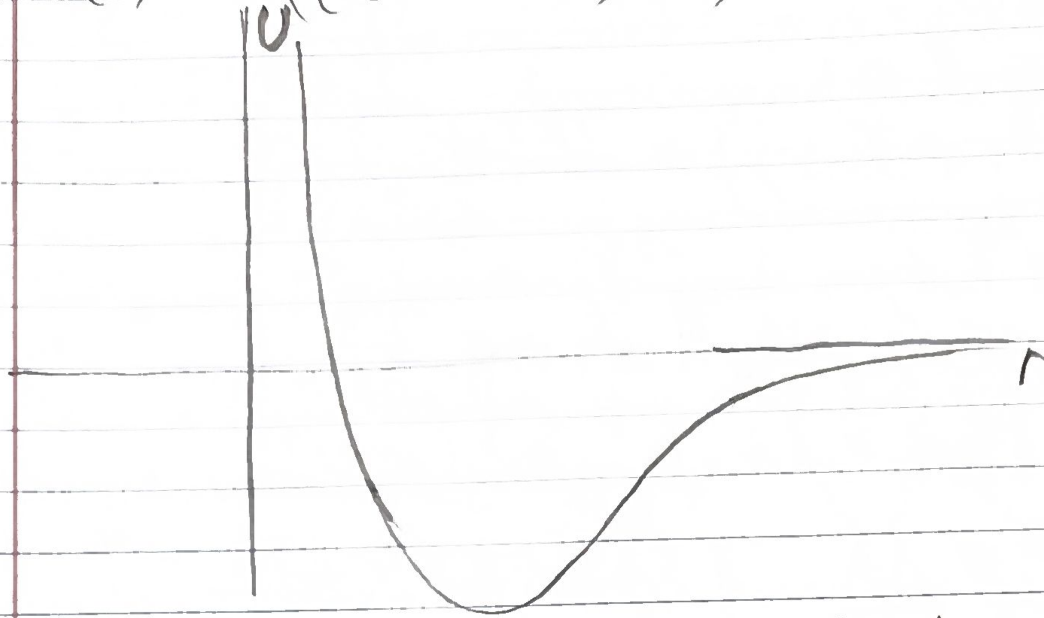
1a) $U(r)=A\left(\left(e^{(R-r)/s}-1\right)^2-1\right)$ $\quad s<<R$



1b) $\frac{dU}{dr}=A\frac{d\left(e^{\frac{(R-r)}{s}}-1\right)^2}{d\left(e^{\frac{(R-r)}{s}}-1\right)}\frac{d\left(e^{\frac{(R-r)}{s}}-1\right)}{dr}$

$=2A\left(e^{\frac{(R-r)}{s}}-1\right)\frac{de^{\frac{R-r}{s}}}{d\left(\frac{R-r}{s}\right)}\frac{d}{dr}\left(\frac{R-r}{s}\right)$

$=-\frac{2A}{s}e^{\frac{(R-r)}{s}}\left(e^{\frac{(R-r)}{s}}-1\right)=0$ $\quad$ No solution for $e^{\frac{R-r^*}{s}}=0$

$e^{\frac{R-r}{s}}=1$ $\quad \frac{R-r}{s}=\ln(1)=0$ $\quad R-r^*=0$

$r^*=R$ $\quad \boxed{r_e=R}$

1c) $R-(R+x)=-x$ $\quad r=R+x$ $\quad x=r-R$

$U(x)=A\left(\left(e^{-x/s}-1\right)^2-1\right)$

$\frac{dU}{dr}=\frac{dU}{dx}\frac{dx}{dr}=\frac{dU}{dx}\frac{d(r-R)}{dr}=\frac{dU}{dx}$

$\frac{dU}{dx}=-\frac{2A}{s}e^{\frac{-x}{s}}\left(e^{-\frac{x}{s}}-1\right)$

But $\left.\frac{dU}{dx}\right|_0=0$ since $x=0$ is a minimum for the potential

Then expanding $U(x)$ in a Taylor series centered at $x=0$,

$$U(x) = U(0) + \frac{1}{2}\frac{d^2U}{dx^2}\Big|_0 x^2$$

$$\frac{d^2U}{dx^2} = \frac{-2A}{s}\frac{d}{dx}\left(e^{-x/s}\left(e^{-\frac{x}{s}}-1\right)\right) = \frac{-2A}{s}\left(\frac{de^{-x/s}}{dx}\left(e^{-\frac{x}{s}}-1\right) + e^{-\frac{x}{s}}\frac{d}{dx}\left(e^{-\frac{x}{s}}-1\right)\right)$$

$$= \frac{-2A}{s}\left(-\frac{1}{s}e^{-\frac{x}{s}}\left(e^{-\frac{x}{s}}-1\right) - \frac{1}{s}e^{-\frac{2x}{s}}\right) \qquad -\frac{1}{s}e^{-\frac{2x}{s}} = -\frac{1}{s}e^{-\frac{x}{s}}\left(e^{-\frac{x}{s}}\right)$$

$$= \frac{2A}{s^2}e^{-\frac{x}{s}}\left(e^{-\frac{x}{s}} - 1 + e^{-\frac{x}{s}}\right) = \frac{2A}{s^2}e^{-\frac{x}{s}}\left(2e^{-\frac{x}{s}} - 1\right)$$

$$\frac{d^2U}{dx^2}\Big|_0 = \frac{2A}{s^2}(2-1) = \frac{2A}{s^2} \qquad U(0) = A\left(\left((1-1)^2 - 1\right)\right) = -A$$

$$U(x) = -A + \frac{A}{s^2}x^2 + \cdots$$

1d) $k = \frac{d^2U}{dx^2}\Big|_0 = \frac{2A}{s^2}$ Effective spring constant when two terms of the expansion are taken. SHO-like motion around $x=0$.

$$U(x) = -A + \frac{1}{2}kx^2 \qquad F = -\frac{dU}{dx} = -\frac{1}{2}k\frac{dx^2}{dx} = -kx$$

$$m\frac{d^2x}{dt^2} = -\frac{2A}{s^2}x \qquad \frac{d^2x}{dt^2} = -\frac{2A}{ms^2}x = -\omega^2 x$$

$$\omega = \sqrt{\frac{2A}{ms^2}} \qquad f = \frac{1}{2\pi}\sqrt{\frac{2A}{ms^2}}$$

2a) $x(t) = A\cos\left(\frac{2\pi}{T}t + \phi\right)$ General solution for SHO

$$\langle x \rangle = \frac{1}{T}\int_0^T A\cos\left(\frac{2\pi}{T}t + \phi\right)dt \qquad u = \frac{2\pi}{T}t + \phi$$

$$\frac{du}{dt} = \frac{2\pi}{T}$$

$$dt = \frac{T}{2\pi}du$$

$$\frac{1}{T}\int_{\phi}^{2\pi+\phi} A\cos(u)\frac{T}{2\pi}du = \frac{A}{2\pi}\int_{\phi}^{2\pi+\phi}\cos(u)du$$

$$= \frac{A}{2\pi}\left[\sin(u)\right]_{\phi}^{2\pi+\phi} = \frac{A}{2\pi}\left(\sin(2\pi+\phi)-\sin(\phi)\right)$$

But $\sin(2\pi+\phi)=\sin(\phi)$ since the period of a sine function is 2π. Returns to the same point after a shift of 2π.

Thus,

$$\langle x\rangle = \frac{A}{2\pi}(0) = 0$$

2b) $$v(t) = \frac{dx}{dt} = -\frac{2\pi A}{T}\sin\left(\frac{2\pi}{T}t+\phi\right)$$

$$u = \frac{2\pi t}{T}+\phi$$

$$\langle v\rangle = \frac{1}{T}\int_0^T -\frac{2\pi A}{T}\sin\left(\frac{2\pi}{T}t+\phi\right)dt \qquad \frac{du}{dt} = \frac{2\pi}{T}$$

$$dt = \frac{T}{2\pi}du$$

$$= -\frac{2\pi A}{T^2}\frac{T}{2\pi}\int_{\phi}^{2\pi+\phi}\sin(u)du = -\frac{A}{T}\int_{\phi}^{2\pi+\phi}\sin(u)du$$

$$= -\frac{A}{T}\left[-\cos(u)\right]_{\phi}^{2\pi+\phi} = \frac{A}{T}\left(\cos(2\pi+\phi)-\cos(\phi)\right)$$

But $\cos(2\pi+\phi)=\cos(\phi)$ since the period of a cosine function 2π. Thus,

$$\langle v\rangle = 0$$

2c) $$K(t) = \frac{1}{2}mv^2(t) = \frac{1}{2}m\left(-\frac{2\pi A}{T}\right)^2\sin\left(\frac{2\pi t}{T}+\phi\right)^2$$

$$= \frac{2\pi^2 mA^2}{T^2}\sin\left(\frac{2\pi t}{T}+\phi\right)^2$$

$$U(t) = \frac{1}{2}kx(t)^2 = \frac{1}{2}k\left(A\cos\left(\frac{2\pi}{T}t+\phi\right)\right)^2$$

$$= \frac{1}{2}kA^2\cos\left(\frac{2\pi t}{T}+\phi\right)^2$$

$$\langle K\rangle = \frac{1}{T}\int_0^T \frac{2\pi^2 MA^2}{T^2}\sin\left(\frac{2\pi t}{T}+\phi\right)^2 dt$$

$u = \frac{2\pi t}{T}+\phi$

$\frac{du}{dt} = \frac{2\pi}{T}$ $\quad dt = \frac{T}{2\pi}du$

$$\frac{T}{2\pi}\int_{\phi}^{2\pi+\phi}\sin(u)^2 du$$

$$\sin^2 u = \frac{1-\cos 2u}{2}$$

$$\int_0^T \sin(\omega t+\phi)^2 dt$$

$$=\frac{T}{2\pi}\frac{1}{2}\int_{\phi}^{2\pi+\phi}(1-\cos 2u)\,du = \frac{T}{4\pi}\left[u-\frac{1}{2}\sin 2u\right]_{\phi}^{2\pi+\phi}$$

$$=\frac{T}{4\pi}\left(2\pi+\phi-\frac{1}{2}\sin(2(2\pi+\phi))-\phi+\frac{1}{2}\sin(2\phi)\right)$$

$$=\frac{T}{4\pi}\left(2\pi+\frac{1}{2}\left(\sin(2\phi)-\sin(2(2\pi+\phi))\right)\right)$$

But $\sin(4\pi+2\phi)=\sin(2\phi)$. Any integer multiple of 2π added to an angle returns to the same value of the function. Then,

$$\frac{T}{4\pi}\left(2\pi+\frac{1}{2}(0)\right)=\frac{T}{2}$$

This implies

$$\left\langle \sin\left(\frac{2\pi t}{T}+\phi\right)^2\right\rangle = \frac{1}{T}\left(\frac{T}{2}\right)=\frac{1}{2}$$

Then,

$$\langle K\rangle = \frac{2\pi^2 MA^2}{T^2}\left(\frac{1}{2}\right)=\frac{\pi^2 MA^2}{T^2}$$

$$\langle U\rangle = \frac{1}{T}\int_0^T \frac{1}{2}kA^2\cos\left(\frac{2\pi}{T}t+\phi\right)^2 dt$$

$u = \frac{2\pi}{T}t+\phi$

$\frac{du}{dt} = \frac{2\pi}{T}$

$dt = \frac{T}{2\pi}du$

$$\int_0^T \cos(\omega t+\phi)^2\,dt$$

$$\frac{T}{2\pi}\int_{\phi}^{2\pi+\phi}\cos(u)^2\,du \qquad \cos^2 u = \frac{1+\cos 2u}{2}$$

$$= \frac{T}{2\pi}\int_{\phi}^{2\pi+\phi}\frac{1+\cos 2u}{2}\,du = \frac{T}{4\pi}\left[u+\frac{1}{2}\sin 2u\right]_{\phi}^{2\pi+\phi}$$

$$= \frac{T}{4\pi}\left(2\pi+\phi+\frac{1}{2}\sin(2(2\pi+\phi))-\phi-\frac{1}{2}\sin(2\phi)\right)$$

$$= \frac{T}{4\pi}\left(2\pi+\frac{1}{2}\left(\sin(2(2\pi+\phi))-\sin(2\phi)\right)\right)$$

But $\sin(4\pi+2\phi)=\sin 2\phi$. Then,

$$\frac{T}{4\pi}(2\pi)=\frac{T}{2}$$

This implies

$$\left\langle \cos\left(\frac{2\pi t}{T}+\phi\right)^2\right\rangle = \frac{1}{T}\left(\frac{T}{2}\right)=\frac{1}{2}$$

Then,

$$\langle U\rangle = \frac{1}{2}kA^2\left(\frac{1}{2}\right)=\frac{1}{4}kA^2$$

We can write k as, using the natural frequency $\omega=\frac{2\pi}{T}$,

$$\omega^2=\frac{k}{m} \qquad k=m\omega^2=m\left(\frac{2\pi}{T}\right)^2$$

to get

$$\langle U\rangle = \frac{1}{4}m\left(\frac{2\pi}{T}\right)^2A^2=\frac{\pi^2 mA^2}{T^2}$$

Therefore,

$$\langle K\rangle = \langle U\rangle$$

All constants π^2, m, A^2, T^2 are non-zero hence

$$\langle K\rangle = \langle U\rangle > 0$$

We can write $\langle K\rangle$ and $\langle U\rangle$ in terms of $\omega = \frac{2\pi}{T}$,

$$\langle K\rangle = \langle U\rangle = \frac{\pi^2 m A^2}{T^2} = \left(\frac{\omega}{2}\right)^2 m A^2 = \frac{1}{4} m\omega^2 A^2$$

But $m\omega^2 = k$,

$$\langle K\rangle = \langle U\rangle = \frac{1}{4} k A^2 = \frac{1}{2}\underbrace{\left(\frac{1}{2} k A^2\right)}_{\text{Total energy } E} = \frac{1}{2} E$$